JDBC Question Answers

1. What is JDBC?

JDBC (Java Database Connectivity) is an API that allows Java applications to interact with databases.
It helps in connecting, executing SQL queries, and retrieving results from databases.

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2. Why is JDBC used?

JDBC is used to:

- Connect Java applications with databases
- Execute SQL queries (SELECT, INSERT, UPDATE, DELETE)
- Retrieve and process data
- Make Java programs database-independent

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3. What are the main components of JDBC?

Main JDBC components are:

1. Driver Manager
2. Driver
3. Connection
4. Statement
5. Result Set
6. SQLException

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4. What is a JDBC Driver?

A JDBC Driver is a software component that enables Java applications to communicate with a database.

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5. Types of JDBC Drivers

There are **4 types** of JDBC drivers:

1. Type 1 – JDBC-ODBC Bridge Driver (deprecated)
2. Type 2 – Native API Driver

3. Type 3 – Network Protocol Driver
4. Type 4 – Thin Driver (Most commonly used)

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6. Which JDBC driver is most commonly used?

Type 4 (Thin Driver)
Because it is platform independent, fast, and does not require extra software.

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7. What is Driver Manager?

- Driver Manager is a class that manages JDBC drivers and establishes a connection with the database.
- getConnection(String URL, String user, String pass): Connection

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8. What is Connection in JDBC?

- A Connection object represents a session between Java application and database.
- It is present in java.sql package
- Collection interface allow java developers to manage the connection, create statement, handle transaction, and close the connection once operation complete.
- Connection c= DriverManager.getConnection(URL, user, pass)

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9. What is Statement?

Statement is an interface used to execute static SQL queries.

10.Types of Statement in JDBC

1. Statement
2. Prepared Statement
3. Callable Statement

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11. Methods of Connection Interface

1.Create Statement() : Statement

This method create a statement object for executing static sql quires

Statement s = c.CreateStatement();

2.Prepare Statement(String query) : Prepare Statement

Its create a object of prepared statement for executing dynamic or parameterized sql query.

3.SetAutoCommit(Boolean b)

It is used to set the commit true or false

4.Close()

After all operations it is mandatory to close open connection.

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12. Difference between Statement and Prepared Statement

Statement	Prepared Statement
• The statement interface is used for general purpose access to database using static SQL statement.	• PreparedStatement extends the use of statement to execute precompiled SQL statement.
• It cannot pass parameters to SQL queries at runtime.	• It can pass parameters to SQL quries at runtime.
• It cannot accept input parameters.	• It is parameterized statement which may or may not accept IN parameters.
• It is prone to SQL injection	• It is resilient to SQL injection
• No binary communication protocol in statement.	• It is used a non-SQL binary protocol
• It can effectively be used to execute DDL commands – CREATE, DROP or ALTER.	• It is ideal for executing DML commands such as SELECT, UPDATE and INSERT.

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13. What is PreparedStatement?

PreparedStatement is used to execute precompiled SQL queries with parameters.

Instead of writing values directly in sql statement we use placeholder(?) and pass the value at runtime with the help of setter method

Example

Insert : Insert into product(id, name) value(?,?);

Update: update product set name = ? where id =?;

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14. Why we use Prepared Statement

1.It prevents SQL Injection (security) when user give some input to perform DML operation Select, Update, Insert

2.Fast Performance

Here, SQL quires compile runs and execute many times with different values

3.Clean Code

For performance sql quires no need for string concatenation easy to read and maintain

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15. What is Callable Statement?

Callable Statement is used to execute stored procedures in the database.

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16. What is ResultSet?

ResultSet is an object that holds the data returned from a SELECT query.

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17. What is execute(), executeQuery(), and executeUpdate()?

1.execute()

- Used to execute any SQL statement
- Returns a Boolean
- It is used for DDL Commands(Insert, Update, Delete)

boolean result = stmt.execute("SELECT * FROM student");

2.executeQuery()

- Used to execute SELECT queries.
- Returns a ResultSet object.

- It is used for DQL commands used only for Select

  ResultSet rs = stmt.executeQuery("SELECT * FROM student");

  3.executeUpdate()

- Used to execute INSERT, UPDATE, DELETE
- Also used for DDL statements (CREATE, DROP, ALTER)
- Returns an int value

int rows = stmt.executeUpdate("UPDATE student SET marks=80 WHERE id=1");

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18. What happens if you don't close a JDBC connection?

- Memory leaks occur
- Database resources get exhausted
- Application performance degrades

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19. What is auto-commit in JDBC?

Auto-commit means each SQL statement is committed automatically after execution.

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20. Why do we set autoCommit(false)?

To control transactions manually and ensure data consistency using commit and rollback.

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21. What is rollback?

Rollback undoes all database changes since the last commit when an error occurs.

22.What is SQL Injection?

SQL Injection is a security attack where malicious SQL code is inserted through user input.

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23. How does JDBC prevent SQL Injection?

By using PreparedStatement with parameterized queries.

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24. What is batch processing?

Batch processing allows multiple SQL statements to be executed in a single batch to improve performance.

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25. When should batch processing be used?

- Bulk insert/update
- Large data processing
- Performance-critical applications

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26. What is SQLException?

SQLException handles database-related errors such as:

- Connection failure
- Invalid SQL
- Constraint violations

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27. How do you handle exceptions in JDBC?

Using:

- try-catch blocks
- finally block to close resources
- or try-with-resources

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28. Explain CRUD operations using JDBC

CRUD stands for:

- **Create** → INSERT
- **Read** → SELECT
- **Update** → UPDATE
- **Delete** → DELETE
  These operations are performed using JDBC Statement or PreparedStatement.

29.What is JDBC API?

JDBC API provides classes and interfaces to connect Java applications with databases.

30.Explain a JDBC Collection steps?

Step 1: Load/Register driver class from driver software

Step 2: Create Connection with databases by URL, User and Password

Step 3: Create Statement by statement / prepared statement or callable statement to send sql quires to databases

Step 4: Execute sql quires by execute()/ executeUpdate()/ executeQuery()

Step 5: Closed Created connection

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31. How does JDBC work in a login application?

1. User enters username & password
2. JDBC connects to database
3. PreparedStatement checks credentials
4. If record exists → login success
5. Else → login failure

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32. What common JDBC mistakes freshers make?

- Not closing connections
- Using Statement instead of PreparedStatement
- Hardcoding credentials
- Not handling exceptions properly
- Ignoring transaction management